

Ramanujan Neuro-Symbolic Mathematics: A Research Program for Dual-Scale Geometry, $K3 \times T^2$ Compactifications, and the Structure of Singular Cascades

Version 2 — position paper; contains no new proven theorems

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Epistemic status. This is a *program proposal*, not a results paper. Every claim below carries a tier label defined in Section 2: **Tier A** (machine-checked), **Tier B** (established literature, cited), **Tier C** (conjecture or heuristic). At the time of writing, the program’s central dictionary contains *no* Tier A entries. Version 2 supersedes and corrects a previous draft whose Lean 4 listings were vacuous; see the Errata in Section 7.

Abstract

We outline a research program that seeks *exact algebraic surrogates* — drawn from q -series, mock modular forms, and the arithmetic of $K3$ surfaces — for the analytic estimates that govern regularized fluid dynamics and related dual-scale geometries. The program couples (i) a heuristic search engine (RAMA) over Eulerian q -series, (ii) shadow completion in the sense of Zwegers, and (iii) formalization in the Lean 4 proof assistant under a strict audit discipline. We state the program’s five central conjectures precisely, grade every claim within a three-tier epistemic system, and attach to each conjecture an explicit falsification protocol and a formal target. This version withdraws the “machine-verified” claims of the previous draft, whose theorem statements were of type **True** and therefore certified nothing. The contribution of the present document is the honest formulation of the conjectures, the audit rules under which any future claim of proof must be certified, and a milestone plan whose outcomes are decidable.

1 The Proposal

For most of its history, rigorous mathematics has proceeded bottom-up: deriving C from A and B . Ramanujan’s working style — documented most starkly in the *Lost Notebook* [14, 2] — was different: identities were recorded as endpoints, with the connective proofs supplied by later generations. We take that style as *design inspiration*, not as evidence for any particular cognitive model (see Section 4), and propose a three-stage pipeline:

1. **The heuristic engine (RAMA).** Mathematical candidate generation is treated as local search over a symbolic space of Eulerian q -series, guided by an energy functional $E = \alpha C + \beta I + \gamma D$ (complexity, inconsistency, distance-to-anchor terms). **Tier C**: the functional is a design choice; no optimality claim is made.
2. **The shadow bridge.** Candidate mock-symmetries are completed to harmonic Maass forms by computing their non-holomorphic period integrals, following Zwegers [15] (e.g. the weight-3/2 shadow $\eta(\tau)^3$ arising in the $K3$ elliptic-genus decomposition). **Tier B** as a technique; **Tier C** for any specific new identity produced by the engine until certified.
3. **The formal lock.** Surviving statements are formalized in Lean 4 under the audit rules of Section 2.

A caveat governs the entire pipeline. *A proof kernel verifies exactly what is stated, and nothing more.* A theorem whose statement is **True** is verified and worthless. The value of stage (iii) is therefore bounded by the strength of the statements we manage to formalize, not by the presence of a green checkmark. The previous draft violated this principle; the present one is organized around it.

2 Epistemic Protocol

Every assertion in this program carries exactly one grade.

Tier	Meaning	Admission criterion
A	Machine-checked	Lean 4 proof; <code>#print axioms</code> clean; no <code>sorry</code>
B	Established	Peer-reviewed literature, cited inline
C	Conjecture / heuristic	Everything else, however plausible

The following rules are binding on all program artifacts.

- R1.** No `axiom` declarations. (The current `DualScale.lean` baseline satisfies this: zero axioms, thirteen audit certificates.)
- R2.** `sorry` marks a target as OPEN. It is never counted as verified, and its presence must be reported.
- R3.** Theorem statements whose type is **True**, or is otherwise trivially inhabited independently of the mathematical content, are prohibited.
- R4.** Every Tier B claim cites its source. Uncited claims default to Tier C.
- R5.** Numerical claims require exact-arithmetic PASS/FAIL certificates over $\mathbb{Q}$, with inputs locked to their literature sources — the discipline already used by the program’s moduli-map checker, which produced PASS(40) for the $(K-s_7, E-s_{7/2})$ pairing and FAIL for a cross-level pairing.

3 The Conjectural Dictionary

The program’s core object is a proposed dictionary between the $\sqrt{\alpha'}$ -regularized dynamics of an incompressible flow on a macroscopic manifold and arithmetic structures carried by a quantum fiber. Each entry below is presented in four parts: the established anchor (**Tier B**), the conjecture (**Tier C**), the formal target, and the falsification test.

3.1 Hydrodynamic limits and Ramanujan’s sums of tails

Anchor (Tier B). Ramanujan’s “sums of tails” identities give exact closed forms for the defect between certain q -series and their partial sums; a modern treatment, including connections to values of L -functions, is Andrews–Jiménez-Urroz–Ono [1]. On the analytic side, each Fourier–Galerkin truncation of the 3D Navier–Stokes system at frequencies $|k| \leq (\alpha')^{-1/2}$ is a finite-dimensional ODE system and is globally regular by classical theory; blow-up criteria for the untruncated system are governed by results of Beale–Kato–Majda type [3].

Conjecture (Tier C).

Conjecture 1 (Hypothesis U). *There exists $C > 0$, independent of α' , such that the enstrophy density of the $\sqrt{\alpha'}$ -truncated flow satisfies $\sup_t \alpha' \cdot \mathcal{E}_{\alpha'}(t) \leq C$ for all sufficiently small $\alpha' > 0$, and the truncation defect of the energy flux admits an exact sums-of-tails representation.*

Remark 1. The substantive content is *uniformity in α'* . Regularity of each truncated system is classical and proves nothing about Navier–Stokes; and even granting Conjecture 1, the passage to the $\alpha' \rightarrow 0$ limit requires a compactness argument that is itself open. No claim of having “solved” or “bypassed” the Navier–Stokes problem is made here, and the corresponding claim in the previous draft is withdrawn.

Falsification. Exhibit, in DNS surrogates, growth of $\alpha' \cdot \mathcal{E}_{\alpha'}$ along a sequence $\alpha' \rightarrow 0$; or prove that the flux-defect series is not of sums-of-tails type.

3.2 Spectral gaps and Ramanujan graphs

Anchor (Tier B). Deligne’s proof of the Ramanujan–Petersson conjecture [8] gives $|\tau(p)| \leq 2p^{11/2}$ for the Ramanujan τ -function. Lubotzky–Phillips–Sarnak [13] used bounds of this type to construct k -regular Ramanujan graphs, whose non-trivial adjacency eigenvalues satisfy $|\lambda| \leq 2\sqrt{k-1}$ — the optimal spectral gap by Alon–Boppana.

Conjecture (Tier C).

Conjecture 2. *The triad-interaction graph of sub-cutoff modes, with edge weights induced by the fiber arithmetic, is asymptotically Ramanujan. Consequently, resonant nonlinear echoes decay at the optimal mixing rate set by the spectral gap.*

Remark 2. Deligne’s bound is a theorem about modular forms; that it *governs* the interaction graph of a truncated flow is precisely what is conjectured, not established. Nothing is “erased” or “weaponized” by citation alone.

Falsification. Compute the adjacency spectra of the induced graphs for small primes $p \in \{2, 3, 5\}$ under Rule R5. A single certified eigenvalue outside the Alon–Boppana window refutes the conjecture in that regime.

3.3 Vortex-direction constraints and elliptic integrals

Anchor (Tier B). Constantin–Fefferman [6] showed for 3D Navier–Stokes that Lipschitz-type control of the vorticity direction $\xi = \omega/|\omega|$ in regions of high vorticity precludes blow-up; Constantin–Fefferman–Majda [7] developed the analogous geometric constraints for 3D Euler.

Conjecture (Tier C).

Conjecture 3. *For the truncated flow, the Lipschitz modulus $\|\nabla \xi\|_\infty$ equals, up to an explicit constant, the truncation angle $\alpha < \pi/2$ of an incomplete elliptic integral attached to the fiber, with the angle determined by the T -dual cutoff.*

Remark 3. At present there exists *no derivation* linking the modular parametrization of incomplete elliptic integrals to the direction field of a Navier–Stokes vortex. The correspondence is proposed on structural grounds (both objects arise as truncations of a scale symmetry) and stands or falls with Milestone M3 of Section 6.

Falsification. Derive the leading-order relation and compare, in exact arithmetic where possible, against the CF criterion constants; a mismatch at leading order refutes the identification.

3.4 Criticality, continued fractions, and the singular set

Anchor (Tier B). Caffarelli–Kohn–Nirenberg [5] proved that for suitable weak solutions of 3D Navier–Stokes the singular set has vanishing one-dimensional parabolic Hausdorff measure; in particular its dimension is at most 1. Separately, the Rogers–Ramanujan continued fraction exhibits genuinely discrete limiting behavior on the unit circle: at certain roots of unity it diverges while possessing finitely many subsequential limits [4, 2].

Conjecture (Tier C).

Conjecture 4. (i) *The singular set of suitable weak solutions has Hausdorff dimension zero.* (ii) *As $\alpha' \rightarrow 0$, the truncated flow selects among finitely many non-communicating topological states indexed by continued-fraction limit classes, rather than diverging to a single blow-up profile.*

Remark 4. Part (i) is strictly *stronger* than the known CKN bound and is open; the previous draft presented it as “formally proved,” which was false. Part (ii) transfers a theorem about special functions to a statement about PDE dynamics; the transfer is the conjecture.

Falsification. Any construction of a singular set of positive dimension refutes (i); numerical continuation exhibiting a continuum of limit states as $\alpha' \rightarrow 0$ refutes (ii).

3.5 $K3$ geometry, Mathieu moonshine, and the $\mathrm{Sym}^2(L_2)$ lock

Anchor (Tier B). The elliptic genus of $K3$ decomposes into characters whose multiplicities are dimensions of representations of the Mathieu group M_{24} (Eguchi–Ooguri–Tachikawa [10]); the existence of the underlying M_{24} -module was proved by Gannon [11]. The mock-modular pieces are completed by shadows following Zwegers [15], and homological blocks connect such mock objects to quantum invariants [12]. For one-parameter families of high Picard rank $K3$ surfaces (e.g. with Shioda–Inose structure), the third-order Picard–Fuchs operator is the symmetric square of a second-order operator [9].

Conjecture (Tier C).

Conjecture 5. *The macroscopic transport operator of the dual-scale system is conjugate to $L_3 = \mathrm{Sym}^2(L_2)$, where L_2 is the Picard–Fuchs operator of the fiber family; the non-holomorphic shadow completes the T -dual layer to a modular object.*

Program note (Tier C, certificate pending). Within Stream 1’s candidate selection over the sporadic sequences (s_7, s_{10}, S_{22}) , the S_{12} sequence is *provisionally* reclassified from $K3$ candidate to elliptic-curve background. Under Rule R5 this classification remains Tier C until its moduli map carries an exact-arithmetic certificate of the same kind already obtained for the $(K-s_7, E-s_{7/2})$ pairing.

Falsification. A certified FAIL of the S_{12} moduli map against the elliptic-curve background, or a certified PASS against a $K3$ family, decides the reclassification either way.

4 Reading the Lost Notebook: Motivation, Not Evidence

The 1988 Narosa facsimile [14] and the Andrews–Berndt volumes [2] organize a body of material with a genuine throughline into the present program:

- The mock theta functions of the manuscript were completed to harmonic Maass forms by Zwegers [15] and later connected to the $K3$ elliptic genus [10, 11]. That historical arc is real (**Tier B**) and motivates Section 3.5; it does not show that Ramanujan was “calculating string compactification dimensions.”
- The continued-fraction material supplies the discrete-limit phenomena (**Tier B**) that motivate Conjecture 4(ii).
- The asymptotic material sits in the lineage of the circle method and motivates the saddle-point entries of the dictionary.
- The absence of connective proofs in the manuscript is a historical observation about Ramanujan’s working style. We adopt it as a *design metaphor* for the RAMA engine. It is not evidence that his cognition implemented the functional $E = \alpha C + \beta I + \gamma D$, and the previous draft’s claim to that effect is withdrawn.

5 Formalization Status

Baseline. The program’s Lean 4 artifact (`DualScale.lean`, rebuilt after the discovery of a vacuously-true axiom derivation in an earlier file) satisfies Rules R1–R3: zero `axiom` declarations and thirteen audit certificates. That baseline is the standard against which this paper’s targets are measured.

Withdrawal. The five listings of the previous draft (`HydrodynamicLimit`, `SpectralGap`, `CFMConstraints`, `PhaseTransitions`, `FiberLock`) all had the form `theorem ... : True := by trivial`. Under Rule R3 they certify nothing and are withdrawn in full.

Honest targets. The correct formal posture for an open conjecture is an explicit, contentful statement marked `sorry`. The following is *schematic* (it does not typecheck as shown; the supporting definitions are Stream 1 work items) and certifies nothing — it declares what would have to be closed for Conjecture 1 to reach Tier A.

Listing 1: Formal target (OPEN). `sorry` = unproven; this listing is a specification, not a verification.

```

1 namespace DualScale.NS
2
3 /-- Truncation scale; exact rational, per audit rule R5. -/
4 def alphaPrime : Rat := 1 / 100
5
6 /-- Enstrophy density of the Galerkin truncation at |k| <= alphaPrime^(-1/2).
7   (Definition to be supplied; specification only.) -/
8 -- def enstrophyDensity (u : TruncatedFlow alphaPrime) (t : Real) : Real := ...
9
10 /-- Hypothesis U, stated with content (uniformity in alphaPrime).
11   OPEN TARGET: the ‘sorry’ below is the honest marker of that fact. -/
12 theorem hypothesisU_uniform_bound :
13   exists C : Real, 0 < C /\
14     forall a : Rat, 0 < a -> a <= alphaPrime ->
15       forall t : Real, a * enstrophyDensity_spec a t <= C := by
16   sorry
17
18 end DualScale.NS

```

Audit note: any future claim that a target above has been closed must be accompanied by a clean `#print axioms` report and the absence of `sorry` in the dependency cone.

6 Milestones

Each milestone has a decidable outcome.

- M1.** Formalize the *statements* of Conjectures 1–5 so that all supporting definitions compile; **sorry** only at the theorem level.
- M2.** Exact-arithmetic spectral certificates (Rule R5) for the triad graphs of Section 3.2 at $p \in \{2, 3, 5\}$: **PASS** inside the Alon–Boppana window or **FAIL** outside it.
- M3.** Leading-order derivation, or refutation, of the elliptic-angle identity of Conjecture 3.
- M4.** Moduli-map certificate deciding the S_{12} reclassification of Section 3.5.
- M5.** Conditional Tier A theorem: Hypothesis U $\Rightarrow$ uniform-in- α' smoothness on $[0, T]$ for the truncated family, with the compactness step stated explicitly as a separate open lemma.

7 Errata to Version 1

1. The author byline “Fields Medal & Nobel Prize in Physics Laureate Framework” is removed. No such distinction exists or is claimed.
2. All five Lean listings are withdrawn as vacuous (**True := by trivial**); no machine verification of any mathematical content occurred.
3. The phrases “Hypothesis U is proven algebraically,” “singularities are geometrically forbidden,” and “zero-axiom, computationally verified bridge” are retracted. Hypothesis U is Conjecture 1; the singular-set claim is Conjecture 4 and is stronger than the known CKN bound [5].
4. The claim that the Lost Notebook “proves that Ramanujan operated precisely as a biological RAMA engine” is retracted; see Section 4.
5. The S_{12} reclassification is downgraded from “formally rejected” to provisional, pending a Rule R5 certificate.

8 Conclusion

The worth of this program will be decided by which of Conjectures 1–5 survive contact with certificates, not by the reach of its vocabulary. Nothing here is currently proven. What the program contributes today is a precise, falsifiable dictionary between regularized hydrodynamics and the arithmetic of mock modular forms, and an audit discipline — no axioms, no vacuous goals, exact-arithmetic certificates — strict enough that when a Tier A entry does appear, it will mean something.

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